ATHANASIOS GOURDOMICHALIS
Java Programms
App3.java

A. Modify the **`factorial`** method of class **App2** (Exercise 2) so that when an **overflow** occurs during the factorial calculation, it returns -1. Additionally, you must not change the return type of the **factorial** method, which is **integer**.

An integer overflow in Java occurs when we try to store a value ***greater*** *than* **Integer.MAX_VALUE** *in an integer (variable of type* ***int****).*

One way to do this is to use a local variable of type **long** during the product calculation, for example:

```
long res_long;
```

If the value of **res_long**, during the factorial calculation, **exceeds** the value of **Integer.MAX_VALUE** the calculation should not continue, and -1 should be assigned to the result, i.e.:

```
int res = -1
```

Otherwise, the calculation continues, and at the end, the calculated result is cast and assigned to the variable **res**, i.e.:

```
res = (int) res_long;
```

Finally, the function returns **`res`**, which is of type **`int`**.

B. Modify the **main method** of class **App2** (Exercise 2) so that in case of an **overflow**, the program displays an **appropriate message and terminates its execution**.

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Modify class App2 based on what is described above (Exercise 3) and submit it as **App3** (file `App3.java`).

Every program must compile and run from the command line (not through an IDE).